

CLACELL: Robust cell type classifier for immune cells

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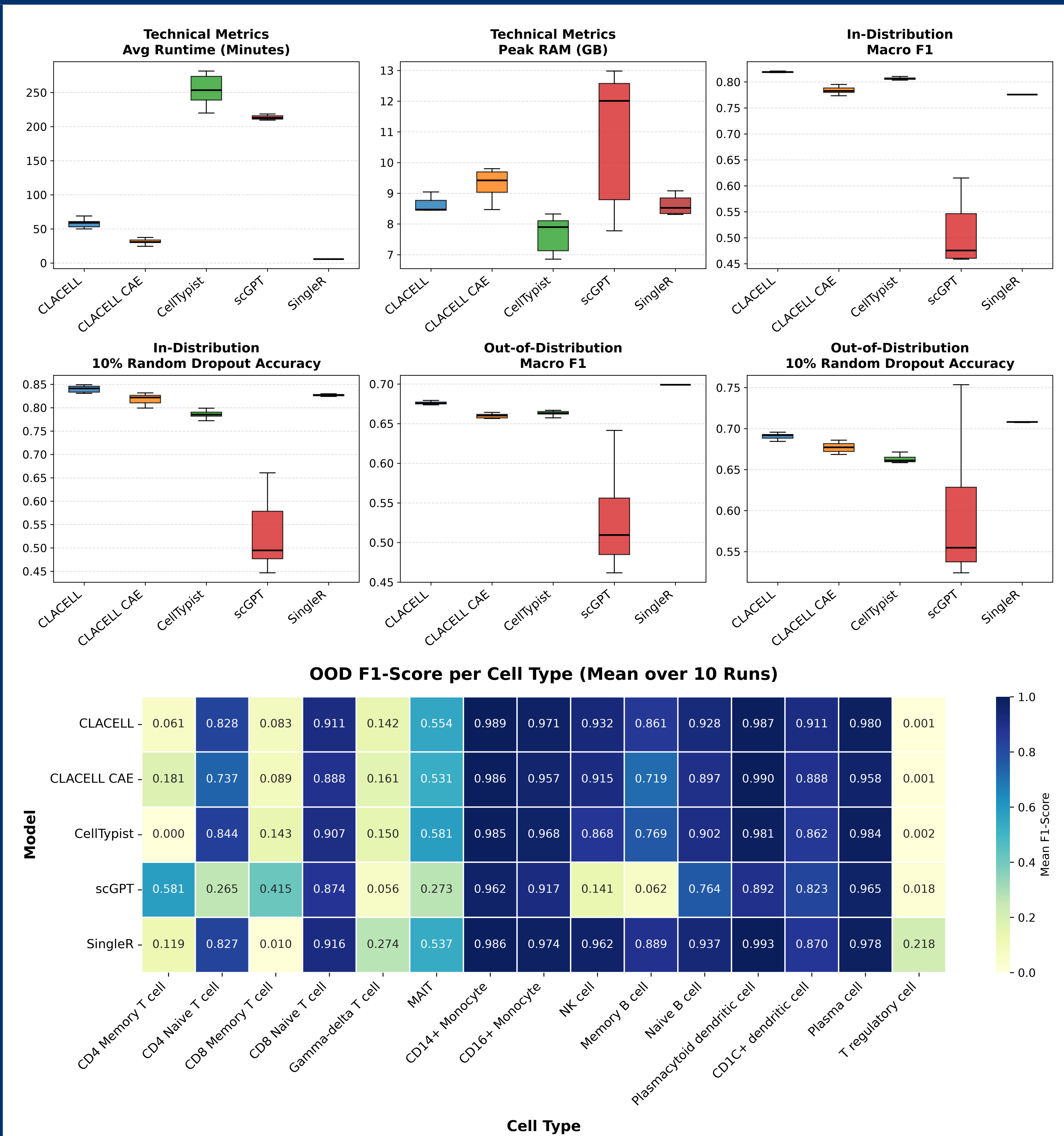
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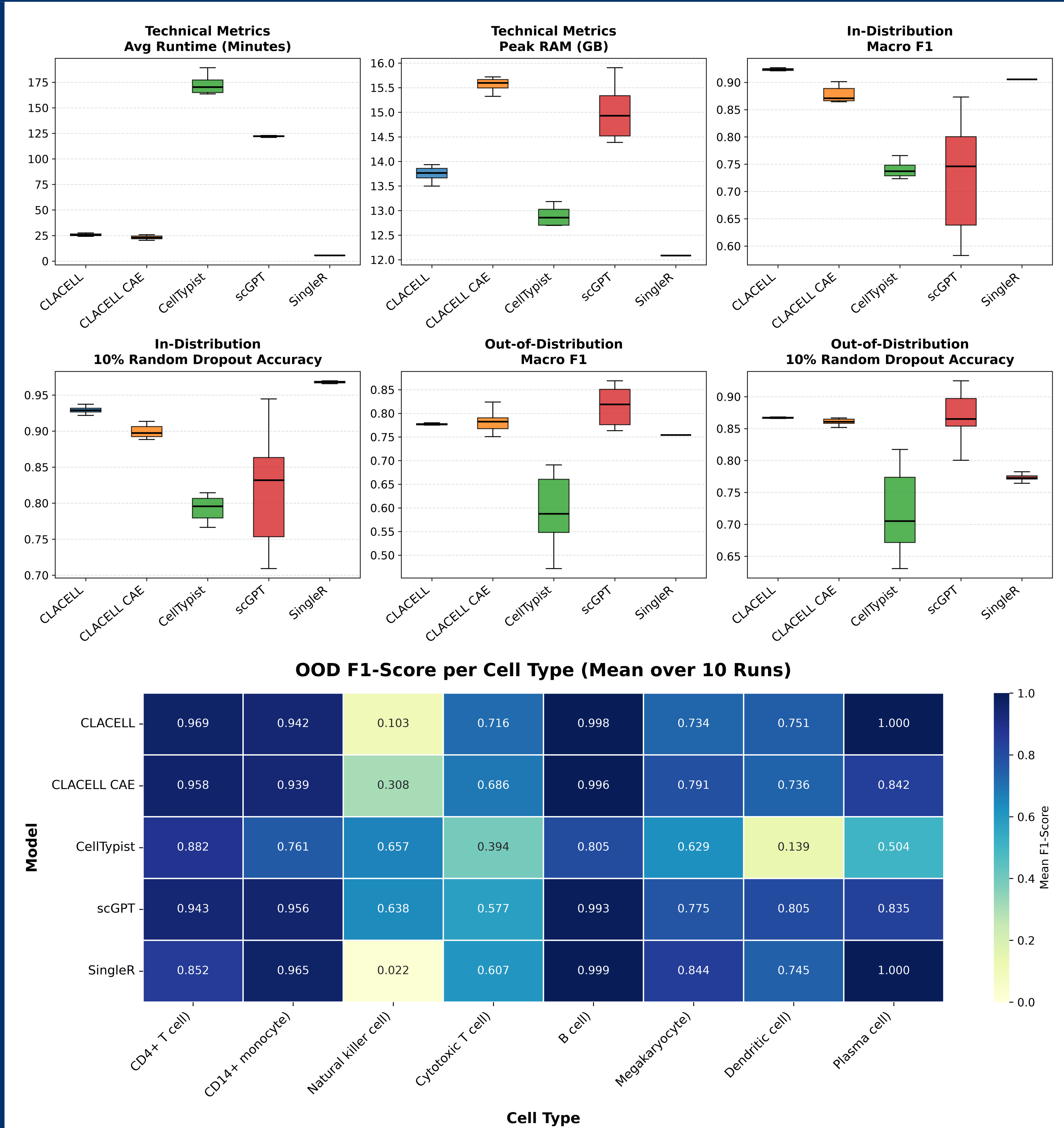
Motivation

- **Background:** Single-cell RNA sequencing (scRNA-seq) reveals cellular heterogeneity, but assigning cell types remains a massive analytical bottleneck.
- **Problem:** Manual marker-based annotation is highly subjective and scale-limited and existing automated classifiers often suffer from high performance variance, sensitivity to gene dropout, and overfitting.
- **Our approach:** We propose CLACELL, an open-source Python framework for robust and deterministic cell type classification of single-cell data.

Results on fine grained labels

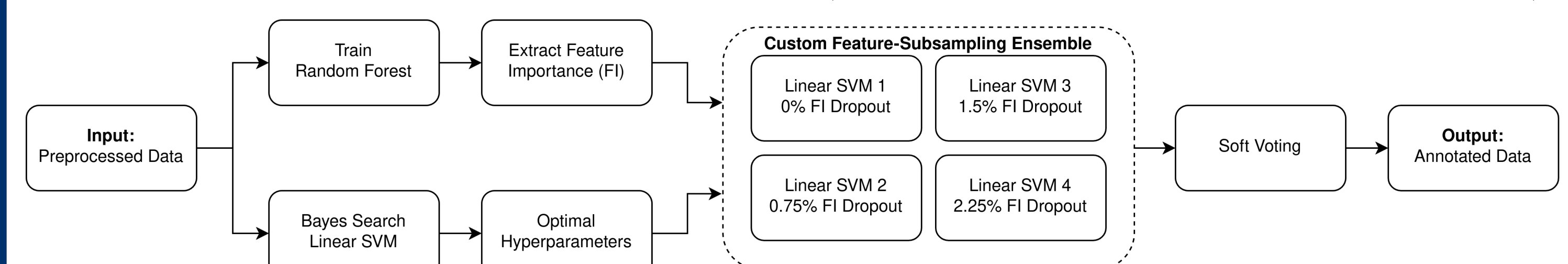


Results on coarse labels



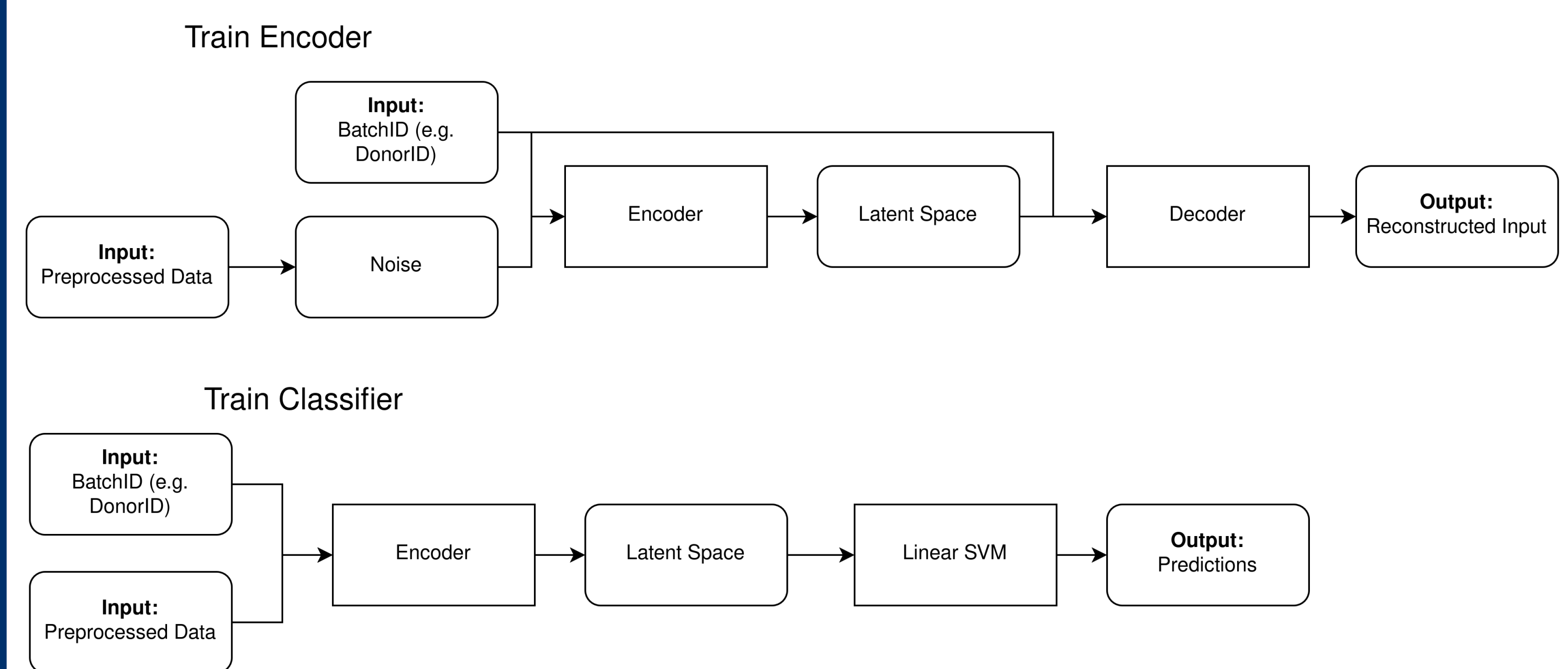
Architecture for the Custom Ensemble

Optimized for high prediction stability (referenced as **CLACELL** in results).



Architecture for the Conditional Autoencoder

Designed for superior Out-of-Distribution (OOD) generalization (referenced as **CLACELL CAE** in results).

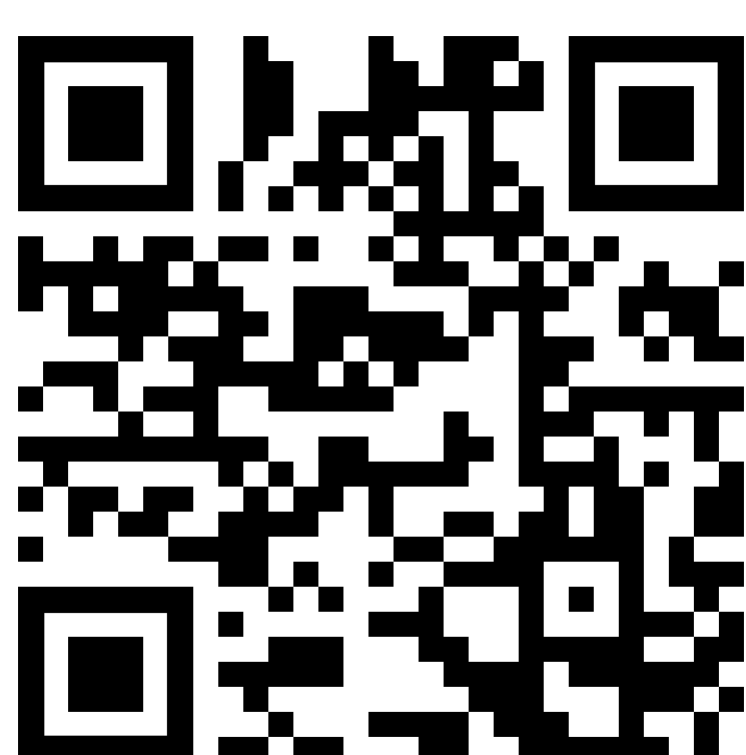


The CLACELL Package

End-to-end Python solution for robust scRNA-seq classification:

- **MarkerAnnotator:** Native Python reimplementation of the *scumi* R algorithm.
- **preprocess_data:** Built-in pipeline (normalization, log-transform, feature selection).
- **CellClassifier:** Custom Ensemble with built-in methods for hyperparameter tuning, training, evaluation, and prediction.
- **ConditionalCellClassifier:** Conditional Autoencoder with built-in methods for hyperparameter tuning, training, evaluation, and prediction.

Github Repository



PyPI Package



Future work

- **Optimization of CLACELL:** Stacking and a combination of Custom Ensemble and Conditional Autoencoder.
- **Hierarchical Classification:** Implement a two-level approach for improved performance.
- **Spatial Data Evaluation:** Assess CLACELL's performance with low gene counts.